SE 2228 ANALYSIS AND DESIGN OF ALGORITHMS
2026 FALL

Prof. Dr. Ahmet KOLTUKSUZ
R.A. Comp. Eng. Mert Hazar HASGÜL

Course Structure

Lecturers

- Prof. Dr. Ahmet KOLTUKSUZ ahmet.koltuksuz@yasar.edu.tr
- R.A. Computer Eng. Mert Hazar HASGÜL mert.hasgul@yasar.edu.tr

Courses

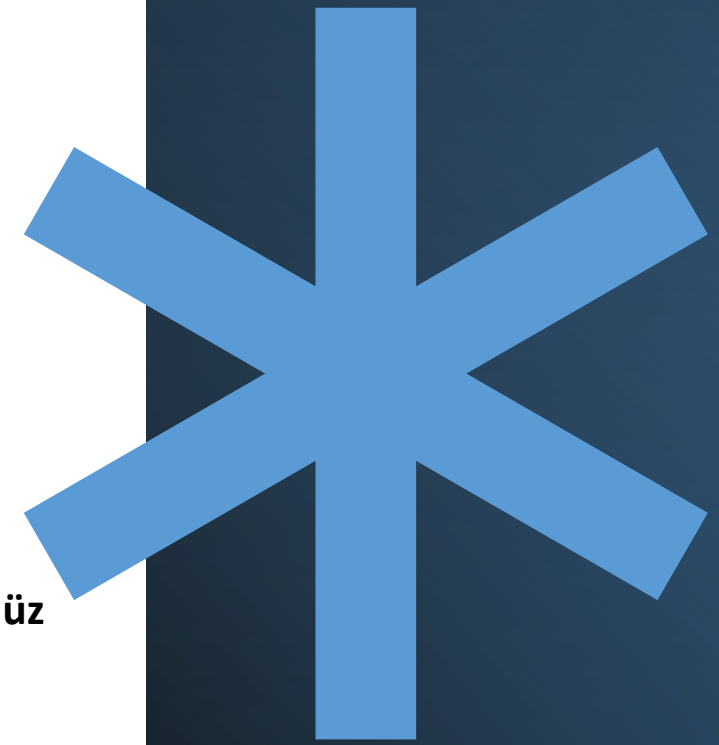
Section #1, 2 & 3 on Fridays, at 10:40-12:30 am @M115
Section #4, 5 & 6 on Fridays, at 02:40-04:30 pm @M116

Labs

Section #1 & 2 on Tuesdays, at 09:40-11:30 am @M433
Section #3 & 4 on Tuesdays at 01:40-03:30 pm @M433
Section #5 & 6 on Thursdays at 12:40-02:30 pm @M434

LMS – Moodle: SE 2228-ALGORİTMA ANALİZİ VE TASARIMI_SB:1-2026-2027Güz
https://online.yasar.edu.tr/course/view.php?id=746

- Announcements
- Lecture notes & Examples
- Anything and everything...



Course Structure



Grading

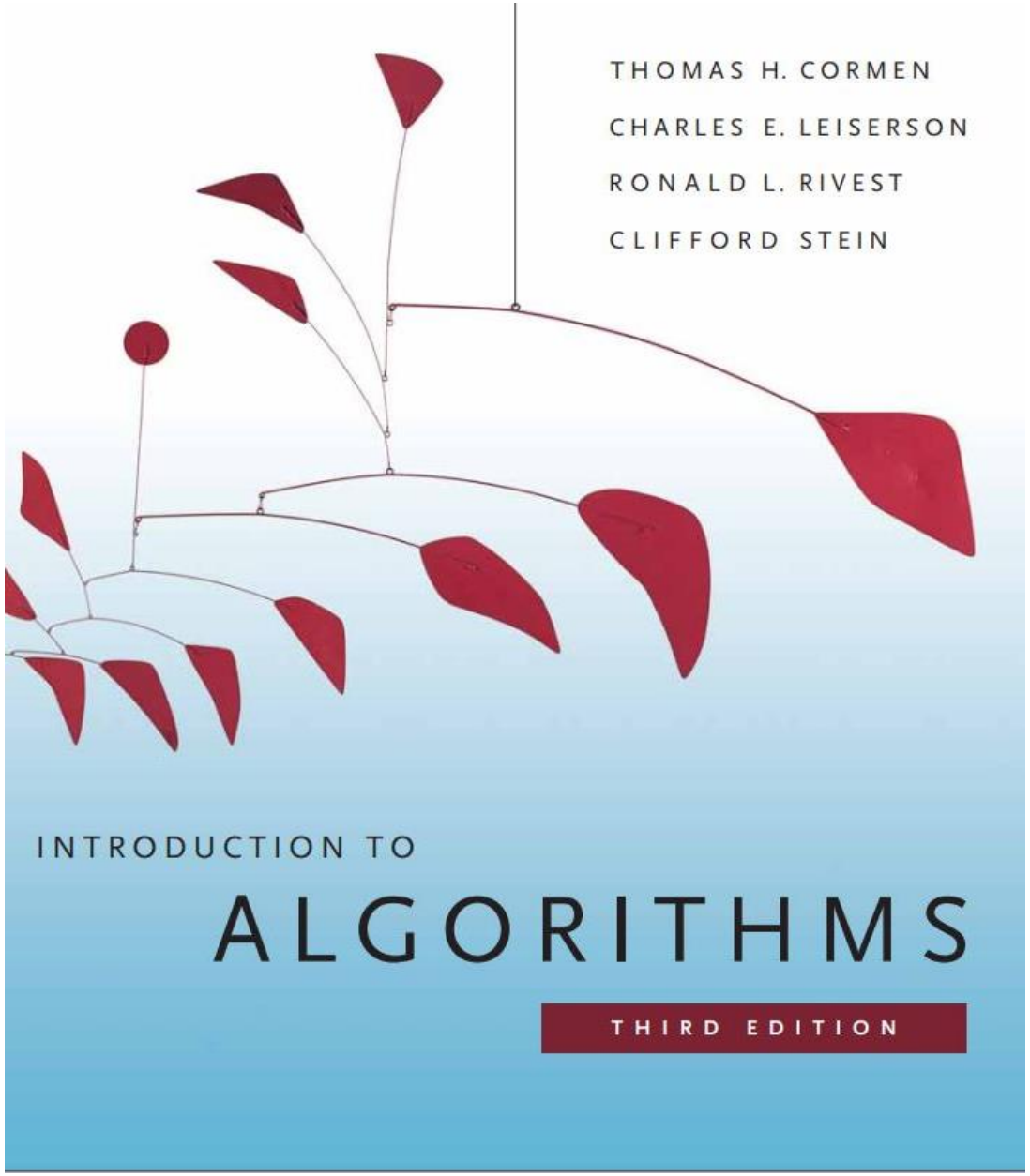
- One Midterm exam with a weight of 40%.
- One Final exam, with a weight of 60%
- The content of the midterm and the final exam will be 50% lab and 50% theory
- **All the exams will be a classical face-to-face type.**

Attendance

- **is compulsory as defined in «Regulation on Associate and Undergraduate Degree Education and Examination» - Ön Lisans ve Lisans Eğitim-Öğretim ve Sınav Yönetmeliği**

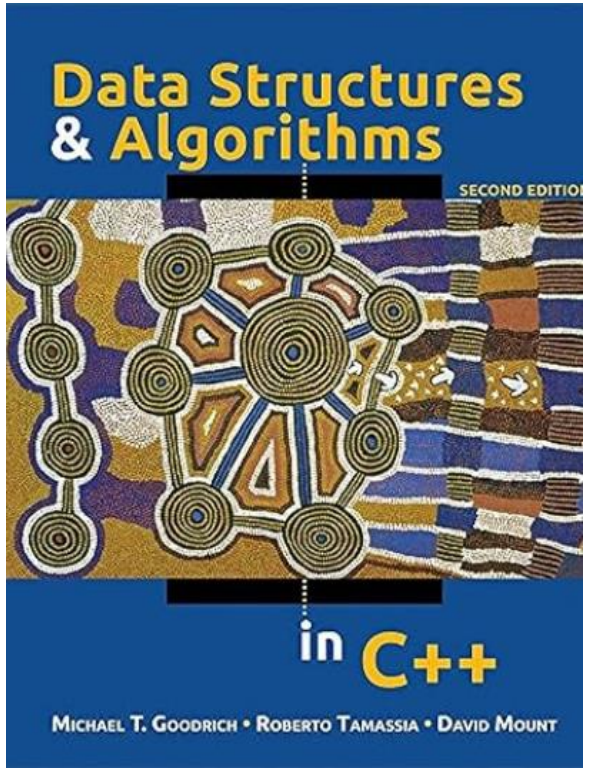
"Introduction to ALGORITHMS", 3rd ed., by Cormen, Leiserson, Rivest, Stein.

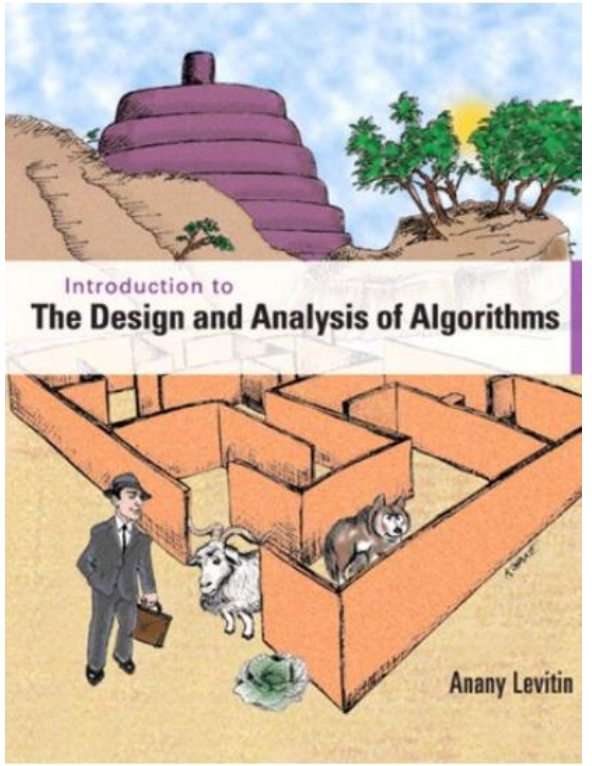
The Course Text Book



Additional Textbooks

- The Design and Analysis of Algorithms, A.Levithin.
- Data Structures and Algorithms in C++,Goodrich,Tamasia,Mount.





Course Description

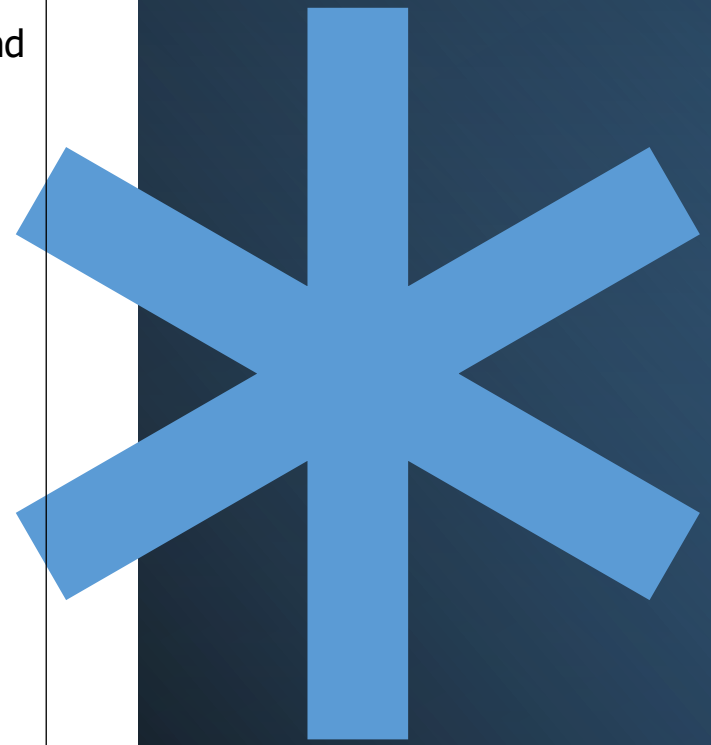
Objectives of the Course

Algorithms play the central role in both the science and practice of computing. Analysis and Design of Algorithms is one of the most fundamental courses in Computer Engineering & Software Engineering curriculum.

The first of the two major objectives of this course is **to introduce students to the fundamental concepts of algorithm analysis and make students apprehend the importance of algorithm efficiency.** Thus, the students will be equipped with knowledge and skills that allow them effectively perform analysis of new algorithms.

The second major objective is **to expose students to algorithm design techniques.** These techniques provide a student with tools for designing algorithms for new problems. In the process of studying algorithm analysis and design several known algorithms will be revised, classified and analysed.

Also, many new algorithms of interest will be considered. This is a third, substantial "by-product" of the course.



Course Description

Learning Outcomes:

1. Students will be able to calculate the complexity of algorithms.
2. Students will be able to use various algorithmic design techniques.
3. Students will be able to design and implement advanced data structures.



The Term Schedule



- There will be one midterm exam at the end of 7th week.
- The final exam will be announced later.
- The course schedule may have minor changes due to updates in the academic schedule or class performance. Should this happen, the class will be informed through the LMS-Moodle right away.

The Weekly Term Schedule

WEEK	THEORY	LAB
01	Introduction, mathematical review, recurrence & solving recurrence relations	C language coding, general coding practices
02	Analysis–1: Basics, Asymptotic Notations, Big O	Summation symbol, logarithms, recursive relations Big O notation, Fibonacci coding
03	Analysis–2: The Other Asymptotic Notations, Iterative & Recursive Algorithms	Fibonacci, linear_binary
04	Design–1: Brute Force Approach	Brute force
05	Design–2: Divide and Conquer Approach	quick sort-merge sort, binary search, matrix multiplication
06	Greedy Algorithms	Dijkstra, Kruskal, Prim
07	Midterm review	Midterm review
08	Heap Structures	Heap Structures
09	Dynamic Programming–1: The Memoization	Floyd-Marshall algorithm
10	Dynamic Programming–2: The Knapsack Algorithm	The knapsack coding
11	Topological Sort	Topological Sort coding
12	Hashing & Digital Signatures	Hash coding, digital signatures
13	Complexity Revisited: P, NP and other classes	Computational Models & alternatives
14	Final Review	Questions & Answers for coding

PLEASE FOCUS ON BELOW ITEMS CAREFULLY

- **In lectures**
  - Do not divide your attention.
  - Do not use your cell phones or laptops.
  - Do not disturb the others in a direct or an indirect way.

- **Take notes on your notebook. Taking photos will not be not allowed.**

- **Emails**
  - Do not send messages via 'OBS’.
  - **Do pay careful attention to your grammar and language when sending emails.**
  - Check your e-mails and course’s Moodle pages regularly.



Wishing you good luck & success!

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